# Doping Stability Report -- Ca in $NaCoO_2$ at 200 C

Generated: 2026-05-30 04:20    Host: $NaCoO_2$    Dopant: Ca

## Executive Summary

Ca preferentially occupies the Na layer. The formation energy difference between the two layers is +0.220 eV, strongly favouring the Na site. MD confirms this preference: on the Na site the dopant MSD plateau slope is 0.00034 A^2/ps (stable), compared to -0.00196 A^2/ps on the Co site.

| Site | E_f (eV) | Charge | Compensation | MSD pass | Coord pass | Verdict |
|---|---|---|---|---|---|---|
| Na | -2.102 | +1 | No | PASS | FAIL | STRUCTURAL COLLAPSE |
| Co | -1.882 | -1 | No | PASS | PASS | STABLE |

## Ca at the Na layer -- STRUCTURAL COLLAPSE

### Charge situation

Ca replaces Na with a charge surplus of +1. CHGNet cannot model this compensation (would require Co oxidation or electron addition). The formation energy below is approximate.

### Formation Energy (thermodynamic stability)

| | | | |
|---|---|---|---|
| Formation energy E_f | **-2.1024 eV** | PASS | *< 1.0 eV to pass* |

E_f = -2.102 eV (approximate -- charge surplus not modelled in CHGNet). This value is strongly negative, indicating that Ca incorporation is thermodynamically very favorable at the Na site.

### Molecular Dynamics Stability (200 C, 25 ps)

| | | | |
|---|---|---|---|
| MSD slope (late-time) | **0.00034 A^2/ps** | PASS | *< 0.005 to pass (plateau = stable)* |
| Final MSD | **0.030 A^2** | ---- | |
| Max displacement | **0.470 A** | ---- | |
| Coordination (min/mean/max) | **3 / 5.7 / 6** | FAIL | *relaxed = 6, tolerance +/-1* |
| Mean bond length (MD vs relaxed) | **2.287 A (relaxed: 2.2** | ---- | |
| Lattice volume change | **+0.00 %** | PASS | *< +/-5% to pass* |

Late-time MSD slope = 0.00034 A^2/ps. The slope is near zero, confirming the dopant is tightly confined to its site. Final MSD = 0.030 A^2; maximum displacement from starting position = 0.47 A.

Coordination fluctuated between 3 and 6 (mean 5.7), deviating by more than +/-1 from the relaxed value (6). This may indicate structural distortion, or that the bond-length cutoff is too close to the actual bond length -- check the coordination_cutoff_A setting. Mean nearest-neighbour distance during MD = 2.287 A.

Volume change = +0.00%. The host lattice is mechanically intact.

**VERDICT: STRUCTURAL COLLAPSE**

The coordination criterion failed while MSD and volume both pass, suggesting a possible cutoff artifact rather than a true structural collapse. The Ca atom barely moves (MSD = 0.030 A^2) but the coordination count fluctuates because the bond length at the Na site is close to the analysis cutoff. Consider increasing coordination_cutoff_A and re-running analysis.

*? WARNING: Charge surplus (+1) cannot be modelled in CHGNet. Running single substitution without compensation. E_f is approximate.*

## Ca at the Co layer -- STABLE

### Charge situation

Ca replaces Co with a charge mismatch of -1. To restore charge neutrality, 1 Na atom(s) were removed from the supercell (furthest from the dopant site). The formation energy accounts for the Na-vacancy term.

### Formation Energy (thermodynamic stability)

| | | | |
|---|---|---|---|
| Formation energy E_f | **-1.8824 eV** | **PASS** | *< 1.0 eV to pass* |

E_f = -1.882 eV. This value is strongly negative, indicating that Ca incorporation is thermodynamically very favorable at the Co site.

### Molecular Dynamics Stability (200 C, 25 ps)

| | | | |
|---|---|---|---|
| MSD slope (late-time) | **-0.00196 A^2/ps** | **PASS** | *< 0.005 to pass (plateau = stable)* |
| Final MSD | **0.056 A^2** | ---- | |
| Max displacement | **0.670 A** | ---- | |
| Coordination (min/mean/max) | **5 / 6.0 / 6** | **PASS** | *relaxed = 6, tolerance +/-1* |
| Mean bond length (MD vs relaxed) | **2.168 A (relaxed: 2.1** | ---- | |
| Lattice volume change | **+0.00 %** | **PASS** | *< +/-5% to pass* |

Late-time MSD slope = -0.00196 A^2/ps. The negative slope is a flat plateau -- essentially no migration detected. Final MSD = 0.056 A^2; maximum displacement from starting position = 0.67 A.

Coordination fluctuated between 5 and 6 (mean 6.0), within the +/-1 tolerance of the relaxed value (6). This is consistent with normal thermal vibration. Mean nearest-neighbour distance during MD = 2.168 A.

Volume change = +0.00%. The host lattice is mechanically intact.

**VERDICT: STABLE**

Ca is thermodynamically favorable and kinetically stable on the Co site at 200 C. All four stability criteria pass: formation energy is favorable, the dopant does not migrate during MD, its coordination environment is preserved, and the host lattice volume is unchanged.

*? WARNING: Charge deficit (-1) cannot be compensated in CHGNet (would require adding Na interstitials or oxidising Co). Running single substitution without compensation. E_f is approximate.*

## Site Preference Comparison

The Na site is preferred by 0.220 eV over the Co site. Formation energy: -2.102 eV (Na) vs -1.882 eV (Co).

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## Caveats and Limitations

? **CHGNet is charge-neutral.**

The potential does not model oxidation states, polaron formation, or electron/hole localisation. Formation energies for aliovalent dopants (charge mismatch != 0) are approximate even when structural compensation defects are included.

? **No Freysoldt/Kumagai correction.**

Electrostatic finite-size corrections for charged defects require the host dielectric tensor and the DFT electrostatic potential, neither of which is available from a neural-network potential. For publication-quality E_f of charged defects, follow up with DFT using doped or pymatgen-analysis-defects.

? **Metal-rich reference state.**

Chemical potentials are computed from elemental ground-state structures. Synthesis under oxide-rich or other conditions shifts all E_f values by additive constants; the relative ordering of sites is unaffected.

? **MD timescales are short (25 ps default).**

This is sufficient to assess local site stability and rapid migration, but cannot resolve slow diffusion mechanisms. A non-plateauing MSD confirms instability; a plateau does not guarantee stability on longer timescales.

? **Energy rankings are more trustworthy than absolute values.**

CHGNet is a universal MLIP trained on r2SCAN DFT. Formation-energy errors for transition-metal oxides are typically 50-150 meV/atom. Use relative orderings (which site is preferred?) as the primary result.